

EEN1095 Project Progress and Change Log

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Project Title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels	Supervisor	Martin Collier
Implementation Period	Week 3 - Week 4	Version Date	05/06/2026

Guidance: Maintain this as one **cumulative** document throughout EEN1095. Add a new entry every two weeks and do not remove earlier records. Progress should be recorded in a timely and concise manner. Where no change has occurred, this should be stated explicitly.

1. Project Overview

Original approved project title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels
Original project objectives	Evaluate three IIR filter structures (Direct-Form, Ladder, WDF) and FIR filters across four Python precision levels (float16, float32, float64, mpmath) using five quantitative metrics to determine whether structure or precision is the primary robustness driver.
Original methodology proposed	Systematic multi-precision comparative evaluation in Python 3.11 using NumPy, SciPy, and mpmath. All filters designed via SciPy with transition bandwidth 0.02. Metrics computed: pole displacement, stability margin, sensitivity norm, frequency response deviation, roundoff noise.
Planned outputs / deliverables	Complete Python multi-precision evaluation framework; experimental results dataset; comparative analysis tables and figures; final project report (August 2026); open-source Python tool for precision-aware filter design.

2. Summary of Key Milestones

The table below is intended as a high-level summary. Use Section 3 for the detailed bi-weekly updates.

Period	Planned activity	Progress made	Outcome / status
Weeks 1-2	Begin Python multi-precision framework development. Set up NumPy, SciPy, and mpmath. Validate precision control for float16, float32, float64, and mpmath.	Python framework structure established. Precision control for float16, float32, float64, and mpmath confirmed working. Pilot filter design tested using SciPy.	In Progress — M3: Python framework validated at all 4 precision levels
Weeks 3-4	Complete Python framework validation. Implement direct-form IIR evaluation module. Run pilot experiments for order-10 and order-20 IIR designs.	Not yet reached. Scheduled for next period.	Upcoming
Weeks 5-6	Implement ladder filter Python module. Validate against direct-form baseline.	Not yet reached. Scheduled for June 2026.	Upcoming — M4: All filter structures implemented and validated
Weeks 7-8	Implement WDF Python module. Begin full IIR experimental evaluation across all 144 configurations.	Not yet reached. Scheduled for June 2026.	Upcoming — M5: Complete experimental dataset collected
Weeks 9-10	Complete full IIR and FIR experimental evaluation. Collect complete results dataset.	Not yet reached. Scheduled for June–July 2026.	Upcoming
Weeks 11-12	Perform cross-precision statistical analysis. Produce comparative tables and figures.	Not yet reached. Scheduled for July 2026.	Upcoming — M6: Analysis and comparative figures complete
Weeks 13-14	Write and submit final project report. Publish open-source Python framework.	Not yet reached. Scheduled for August 2026.	Upcoming — M7: Final report submitted

3. Detailed Progress Entries

Entry 1: Weeks 1-2

Main objectives for this period	Begin Python multi-precision framework development using NumPy, SciPy, and mpmath. Set up and validate precision control for float16, float32, float64, and mpmath. Run pilot filter design tests using SciPy.
Work completed	Python framework structure established. NumPy dtype selection confirmed working for float16, float32, and float64. mpmath precision control via mp.dps = 50 confirmed working. Pilot filter design tested using SciPy scipy.signal for Butterworth and elliptic IIR designs. SciPy SOS format validated for numerical accuracy in high-order IIR designs.
Key outputs produced	Python framework files: filter_design.py, core filter design and precision conversion module. metrics.py, five performance metrics module. experiment.py, experimental evaluation runner. Pilot results for order-10 and order-20 direct-form IIR designs across all four precision levels.
Problems encountered	No significant problems encountered. Minor issue with mpmath coefficient conversion from SciPy float64 arrays, resolved by converting each coefficient individually using mpmath.mpf(str(c)) to avoid floating-point representation errors during conversion.
Changes made to plan or methodology	No changes to plan or methodology at this stage. Framework implementation proceeding as planned.
Reasons for any changes	No changes required. Python framework structure and precision control confirmed working as originally planned.
Next steps	Weeks 3-4 — complete full Python framework validation across all four precision levels. Implement direct-form IIR evaluation module. Run pilot experiments for order-10 and order-20 IIR designs across all precision levels and record results.

Entry 2: Weeks 3-4

Main objectives for this period	Formalize the mathematical definition of filter sensitivity across all five-performance metrics. Implement and run the complete Direct-Form IIR baseline evaluation across all four Python precision levels. Establish the Direct-Form results as the quantitative baseline for comparison against the Ladder filter structure. Set up Git repository and push all framework files to GitHub. Begin studying the Lossless Discrete Integrator transformation as the mathematical foundation for the Ladder filter implementation.
Work completed	Formally defined all five sensitivity metrics — pole

	displacement, stability margin, sensitivity norm, frequency response deviation, and roundoff noise — and confirmed their theoretical basis from Mullis and Roberts 1976 and Ko and Tsai 2020. Implemented <code>direct_form_evaluation.py</code> covering all four filter types (Butterworth, Chebyshev I, Chebyshev II, Elliptic) at orders 10, 20, and 30 across float16, float32, float64, and mpmath. Ran the complete evaluation locally on Python 3.13 and confirmed results. Created Git repository on GitHub, connected via SSH key authentication, and pushed all project files in two commits. Studied the Lossless Discrete Integrator (LDI) transformation from s-plane to z-plane and understood how it preserves the lossless property of analog ladder prototypes. Read and reviewed Ko and Tsai 2020 as the most directly related recent paper to the research question.
Key outputs produced	<code>src/direct_form_evaluation.py</code> — Complete Direct-Form IIR baseline evaluation module covering all filter types, orders, and precision levels with all 5 metrics. <code>src/filter_design.py</code>, <code>metrics.py</code>, <code>experiment.py</code> — Updated with comments and documentation. <code>results/results.json</code> — Pilot results dataset. GitHub repository — https://github.com/darshanmahadevanaika2-sudo/EEN1095-Low-Sensitivity-Digital-Filters — two commits pushed. Literature log updated with Ko and Tsai 2020 as Article Review 2 for Weeks 3-4.
Problems encountered	Problem 1: <code>ModuleNotFoundError</code> — VS Code was using Python 3.14 but NumPy, SciPy, and mpmath were installed on Python 3.13. Resolved by running the script directly using the correct Python 3.13 executable path in Command Prompt. Problem 2: GitHub authentication via password and curl command failed in PowerShell. Resolved by setting up SSH key authentication — generated ed25519 SSH key, added public key to GitHub account, and connected repository via SSH remote URL. Problem 3: PowerShell curl command syntax conflict. Resolved by using <code>Invoke-RestMethod</code> PowerShell equivalent command.
Changes made to plan or methodology	Project scope refined — Wave Digital Filter (WDF) removed from the evaluation scope. Project now focuses exclusively on Direct-Form IIR as the baseline structure and Ladder Filter as the primary investigation structure. This makes the comparison more focused and achievable within the project timeline.
Reasons for any changes	Removing WDF from scope allows deeper and more thorough investigation of the Ladder filter structure specifically. The Ladder filter based on the Lossless Discrete Integrator transformation is the most directly relevant low-sensitivity structure for the research question. A focused two-structure comparison — Direct-Form vs Ladder — produces cleaner and more interpretable results within the available timeframe.

Next steps	Weeks 5-6: Implement the Ladder filter Python module based on Bruton 1975 using the Lossless Discrete Integrator transformation. Derive reflection coefficients α from the normalised analog low-pass prototype. Implement the single ladder section difference equation $y[n] = x[n] + \alpha \times (x[n-1] + y[n-1])$. Validate Ladder filter impulse response and frequency response against Direct-Form baseline at float64. Begin cross-precision comparison of Ladder vs Direct-Form at float16, float32, float64, and mpmath. Push all new code to GitHub with meaningful commit messages.
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Entry 3: Weeks 5-6

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 4: Weeks 7-8

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 5: Weeks 9-10

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]

Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 6: Weeks 11-12

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 7: Weeks 13-14

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]